

TDV 2215 Extended Mode - Complete ESC and CSI Sequence Reference

Overview

When the TDV 2215 Extended Control switch (4.2.12) is ON, the terminal supports a comprehensive set of escape sequences beyond the basic TDV 2115 compatibility mode. These sequences follow ECMA-6, ECMA-35, and ECMA-48 standards where applicable, with TDV 2215-specific extensions.

Activation: ESC Q enables extended mode from TDV 2115 compatibility mode.

Mode Compatibility

TDV 2115 Compatible Mode (Extended Control = OFF)

- Only C0 control codes (00-1F) accepted
- ESC sequences ignored except **ESC Q** (enables extended mode)
- Sequence behavior: ESC ignored, remaining characters displayed

Extended Mode (Extended Control = ON)

- Full C0 control set
 - C1 control codes via ESC sequences
 - CSI sequences with parameters
 - Three-character ESC sequences
 - Device control strings (DCS)
-

C0 Control Characters (00-1F)

TDV 2115 Compatible Codes (Available in both modes)

Code	Hex	Mnemonic	TDV 2115 Function	VT100 Compatible
06	06	ACK	Light 2 on keyboard comes on	No - TDV specific
07	07	BEL	Bell	Yes
08	08	BS	Backspace	Yes
09	09	HT	Horizontal Tab	Extended mode only
0A	0A	LF	Line Feed	Yes
0B	0B	VT	Cursor Down	Different from VT100
0C	0C	FF	Roll Up	No - TDV specific
0D	0D	CR	Cursor Return	Yes
0E	0E	SO	Underline/Attribute start	Different from VT100
0F	0F	SI	Normal/Attribute end	Different from VT100
10	10	DLE	Cursor Load (direct addressing)	No - TDV specific
15	15	NAK	Light 3 on keyboard comes on	No - TDV specific
16	16	SYN	Keyboard lights are turned off	No - TDV specific
17	17	ETB	Roll Down	No - TDV specific
18	18	CAN	Cursor Right	Different from VT100
19	19	EM	Erase Page	No - TDV specific
1A	1A	EOT	Erase Line	Different from VT100
1C	1C	FS	Cursor Up	No - TDV specific
1D	1D	GS	Cursor Home	No - TDV specific
1E	1E	STX	Video Off	No - TDV specific
1F	1F	ETX	Video On	No - TDV specific

Extended Mode Additional C0 Codes

Code	Hex	Mnemonic	Function	VT100 Compatible
09	09	HT	Horizontal Tab	Yes
1B	1B	ESC	Lead-in for additional controls	Yes

C1 Control Codes (ESC 40-5F)

Available only in Extended Mode. Represented as ESC + character in range 40-5F.

Sequence	Hex	C1 Name	Function	VT100 Compatible
ESC [	1B 5B	CSI	Control Sequence Introducer	Yes
ESC ^	1B 5E	PM	Privacy Message	Partial
ESC N	1B 4E	SS2	Single Shift 2	No - Character set
ESC O	1B 4F	SS3	Single Shift 3	No - Character set
ESC P	1B 50	DCS	Device Control String (PUSH-key loading)	Partial
ESC \	1B 5C	ST	String Terminator	Yes
ESC H	1B 48	HTS	Horizontal Tab Set	Yes

Three-Character ESC Sequences (ESC 20-2F 30-3F)

Extended mode only. Format: ESC + intermediate + final character.

Sequence	Function	VT100 Compatible
ESC # 3	DWL - Double-width line (top half)	Yes
ESC # 4	DWL - Double-width line (bottom half)	Yes
ESC # 5	SWL - Single-width line	Yes
ESC # 6	DWL - Double-width line (single height)	Yes

CSI Sequences (ESC [Pn...Pn final)

Control sequences with numeric parameters. Format: ESC [[parameters] final_character

Parameters are ASCII numbers separated by semicolons (;). Omitted parameters use default values.

Cursor Movement

Sequence	Parameters	Function	Default	VT100 Compatible
ESC [Pn A	Pn = count	CUU - Cursor Up	1	Yes
ESC [Pn B	Pn = count	CUD - Cursor Down	1	Yes
ESC [Pn C	Pn = count	CUF - Cursor Forward	1	Yes
ESC [Pn D	Pn = count	CUB - Cursor Backward	1	Yes
ESC [Pl ; Pc H	Pl=line, Pc=column	CUP - Cursor Position	1;1	Yes
ESC [Pl ; Pc f	Pl=line, Pc=column	HVP - Horizontal/Vertical Position	1;1	Yes

Tabulation

Sequence	Parameters	Function	Default	VT100 Compatible
ESC [Pn Z	Pn = count	CBT - Cursor Backward Tabulation	1	Yes
ESC [Ps g	Ps = type	TBC - Tabulation Clear	0	Yes

TBC Parameters:

- Ps = 0: Clear tab at current position
- Ps = 3: Clear all tabs

Editing Functions

Sequence	Parameters	Function	Default	VT100 Compatible
ESC [Pn @	Pn = count	ICH - Insert Character	1	Yes
ESC [Pn P	Pn = count	DCH - Delete Character	1	Yes
ESC [Pn L	Pn = count	IL - Insert Line	1	Yes
ESC [Pn M	Pn = count	DL - Delete Line	1	Yes
ESC [Pn X	Pn = count	ECH - Erase Character	1	Yes

Erase Functions

Sequence	Parameters	Function	Default	VT100 Compatible
ESC [Ps J	Ps = mode	ED - Erase in Display	0	Yes
ESC [Ps K	Ps = mode	EL - Erase Line	0	Yes

ED Parameters:

- Ps = 0: Erase from cursor to end of display
- Ps = 1: Erase from start of display to cursor
- Ps = 2: Erase entire display

EL Parameters:

- Ps = 0: Erase from cursor to end of line
- Ps = 1: Erase from start of line to cursor
- Ps = 2: Erase entire line

Graphic Rendition

Sequence	Parameters	Function	Default	VT100 Compatible
ESC [Ps...Ps m	Ps = attributes	SGR - Select Graphic Rendition	0	Yes

SGR Parameters (can be combined with semicolons):

- 0: Normal (reset all attributes)
- 1: Bold/High intensity
- 2: Low intensity
- 4: Underline
- 5: Blink
- 7: Inverse video
- 8: Invisible (concealed)

Printer Control

Sequence	Parameters	Function	Default	VT100 Compatible
ESC [Ps i	Ps = mode	MC - Media Copy	0	Partial

MC Parameters:

- Ps = 0: Print screen
- Ps = 4: Turn off printer pass-through
- Ps = 5: Turn on printer pass-through

Mode Setting

Sequence	Parameters	Function	Default	VT100 Compatible
ESC [Ps...Ps h	Ps = mode numbers	SM - Set Mode	-	Partial
ESC [Ps...Ps l	Ps = mode numbers	RM - Reset Mode	-	Partial

Mode Numbers:

- 20: LNM - Line Feed/New Line Mode

Device Status Report (DSR) - Host Response Sequences

The TDV 2215 supports device status reporting as documented in the specification.

Status Query

Sequence	Function	Response Format	VT100 Compatible
ESC [5 n	DSR - Device Status Report	ESC [0 n (OK) or ESC [3 n (Error)	Yes
ESC [6 n	CPR - Cursor Position Report	ESC [Pl ; Pc R	Yes

Response Sequences (Terminal → Host)

Response	Format	Description	Notes
ESC [0 n	Device Status	Terminal OK	Standard VT100 response
ESC [3 n	Device Status	Terminal has malfunction	Standard VT100 response
ESC [PI ; Pc R	Cursor Position	Current line (PI) and column (Pc)	Line/column are 1-based

Implementation Notes:

- Cursor position report shows current cursor location
- Line and column numbers start from 1 (not 0)
- Error status reflects hardware self-test results
- Firmware includes "Cursor position:" string handling

DCS Sequences (Device Control String)

Used for PUSH-key programming from the host.

PUSH-Key Programming

Sequence	Function	Format
ESC P Ps ST	Program PUSH-key	ESC P key_number string ESC \

Parameters:

- Ps = PUSH-key number (0-15: 0-7 normal, 8-15 shifted)
- string = sequence to store (text and/or control characters)
- ST = String Terminator (ESC)

Example:

- `ESC P 0 Hello World ESC \` - Programs PUSH-key 1 (normal) with "Hello World"
- `ESC P 8 ESC [ H ESC \` - Programs PUSH-key 1 (shifted) with cursor home sequence

Keyboard Function Key Sequences

When Extended Control is ON, special keys generate escape sequences. The TDV 2215 firmware includes keyboard character normalization and transmission handling.

Control and Navigation Keys

Key	TDV 2115 Mode	Extended Mode	VT100 Compatible
Cursor Up	Cursor Up (1C)	ESC [A	Partial
Cursor Down	Cursor Down (0B)	ESC [B	Partial
Cursor Right	Cursor Right (18)	ESC [C	Partial
Cursor Left	Backspace (08)	ESC [D	Partial
Home	Cursor Home (1D)	ESC [H	Yes

Special Function Keys

Key	Function	Extended Mode	Notes
MODE	Setup access	No sequence	Local function only
PRINT	Print control	Depends on printer mode	May send MC sequence
LINE	Online control	No sequence	Local modem control
CLEAR	Clear lamps	No sequence	Local function

Debugging and Diagnostic Features

Transparent Mode

When Send-Receive Mode switch (4.3.1) = TRANSPARENT, the terminal enters debug mode:

Character Display Mapping:

Code Range	Display Format	Description
00	Blank space	Null character
01-1F	Hex values rotated 90°	Control codes visible
20-7F	Standard ASCII	Normal text
80-FF	Underlined characters	High bit set indication

Behavior:

- **Received data:** Displayed directly with above mapping
- **Transmitted data:** Shown in inverse video/low intensity
- **Control decoding:** Disabled (no sequence processing)
- **Exit method:** MODE key twice to access setup menu
- **Use case:** System programmer debugging aid

Error Reporting and Status

Hardware Self-Test:

- Power-on diagnostic sequence

- Error codes displayed on malfunction
- ERROR lamp indication on keyboard
- Status available via ESC [5 n query

Common Error Conditions:

- NVRAM corruption or failure
- Z80SIO communication errors
- Display controller problems
- Keyboard communication failure

Character Set and Encoding

256-Character Support:

- **Standard ASCII:** 0x20-0x7F (96 characters)
- **Semigraphic Set:** Line drawing, histogram, plot characters
- **Subscript/Superscript:** Numeric formatting characters
- **Control Display:** Visible representation in transparent mode

Character Generator:

- **Matrix:** 9×14 dot cell (effective 7×9 character matrix)
- **Anti-glare:** Bonded anti-reflex faceplate
- **Adjustable:** Height, tilt, swivel stand

Communication Diagnostics

Transmission Monitoring:

- Character-by-character transmission tracking
- Flow control status (XON/XOFF states)
- Buffer utilization monitoring
- Error detection and recovery

Interface Testing:

- Keyboard to host transmission test: `test_keyboard_to_host_transmission()`
- Keyboard communication test: `test_keyboard_communication()`
- Z80SIO port testing: `test_z80sio_ports()`
- Printer status checking: `check_printer_status()`

PUSH-Keys (Programmable Function Keys)

Key	Normal Mode	Shifted Mode	Description
PUSH 1	User programmable	User programmable	12 bytes max sequence
PUSH 2	User programmable	User programmable	12 bytes max sequence
PUSH 3	User programmable	User programmable	12 bytes max sequence
PUSH 4	User programmable	User programmable	12 bytes max sequence
PUSH 5	User programmable	User programmable	12 bytes max sequence
PUSH 6	User programmable	User programmable	12 bytes max sequence
PUSH 7	User programmable	User programmable	12 bytes max sequence
PUSH 8	User programmable	User programmable	12 bytes max sequence

Programming Methods:

1. **Local:** Via setup menu (MODE key twice + SHIFT) if PKP switch = ALLOWED
2. **Remote:** Via DCS sequences from host (always available in extended mode)

Storage: NVRAM addresses 0x60-0x17B (16 sequences × 12 bytes each)

TDV 2215 Specific Extensions

Non-VT100 Compatible Features

1. **TDV 2115 Control Codes:** Unique functions like cursor load (DLE), video control, keyboard lights
2. **Roll Operations:** Roll up/down with smooth scroll option
3. **Attribute Mode:** SO Y SI sequences for graphic rendition
4. **PUSH-Key Programming:** 16 programmable sequences via DCS
5. **Printer Modes:** Local/remote/log modes with specific control
6. **Double-Width Characters:** Line-by-line basis vs character-by-character

VT100 Compatibility Notes

- **Basic cursor movement:** Compatible
- **Character attributes:** SGR sequences compatible, but TDV also supports attribute mode
- **Editing functions:** Standard CSI sequences supported
- **Tab handling:** Standard tab set/clear functions
- **Device status:** Standard DSR responses

Extended Capabilities

- **256 Character Set:** Includes semigraphic characters (line drawing, histogram, subscript, superscript)

- **Soft-Switch Control:** Remote configuration via SM/RM sequences
 - **Background Printing:** Optional print buffer with MC control
 - **Advanced Flow Control:** XON/XOFF and DTR handshaking
-

Implementation Notes

Character Set Support

- **G0:** Primary character set (ASCII + semigraphics)
- **G1:** Alternate character set (semigraphics)
- **SS2/SS3:** Single shift to G2/G3 (limited support)

Graphic Rendition Modes

1. **ATTR Mode:** TDV 2115 style attributes occupy screen positions
2. **UNDERLINE Mode:** SO/SI control individual characters
3. **SGR Mode:** Full CSI m sequence support

Advanced Features and Implementation Details

Scrolling and Roll Operations

The TDV 2215 implements sophisticated scrolling with both TDV 2115 compatibility and extended features:

Function	Implementation	VT100 Compatible
ROLL UP (FF/0C)	<code>scroll_display_region_up()</code>	No - TDV specific
ROLL DOWN (ETB/17)	<code>scroll_display_region_down()</code>	No - TDV specific
Smooth Scroll	Hardware line-by-line scrolling	No - TDV specific
Step Scroll	Text line-by-line scrolling	Partial

Roll Type Control:

- **STEP:** One text line at a time (stepwise)
- **SMOOTH:** One matrix dot line at a time (continuous)
- Controlled by Roll Type switch (4.2.9)

Tabulation System

Advanced tab handling with NVRAM storage:

Function	Implementation	Storage
Default tabs	8-column intervals	Built-in
Custom tabs	User-definable positions	NVRAM
Tab advance	<code>cursor_tab_advance()</code>	Hardware
Tab buffer	Circular buffer system	RAM

Tabulation Functions:

- `get_default_tab_stop_count()`: Returns default tab spacing
- `get_next_tab_stop_from_buffer()`: Retrieves custom tab positions
- Supports both permanent and temporary tab rack settings

Character Attribute System

Three distinct graphic rendition modes:

1. ATTR Mode (TDV 2115 Compatible)

```
SO Y SI - Where Y encodes attributes:  
bit 6,5: 00=control, 01=low intensity, 10=inverse, 11=normal  
bit 4:    0=primary function, 1=alternate (blink/underline/invisible)  
bits 3-0: Additional control/positioning
```

2. UNDERLINE Mode (TDV 2115 Compatible)

```
SO - Start underline/special rendition  
SI - Return to normal rendition
```

3. SGR Mode (Extended - VT100 Compatible)

```
CSI Ps m - Select Graphic Rendition  
Multiple parameters supported with semicolon separation
```

Implementation:

- `calculate_character_attribute_code()`: Computes display attributes
- `process_character_with_attributes()`: Applies rendering
- `set_attribute_processing_mode()`: Switches between modes

Control Character Dispatch System

The firmware uses a sophisticated dual dispatch system:

assembly

```
; Control Character Dispatcher (0x25A6)
MOV B,A          ; Save character
ANI 0x1F         ; Mask to 5 bits (C0 range)
ADD A           ; Index * 2 (16-bit pointers)
; Jump table lookup and call

; High Character Dispatcher (0x25A0)
MOV B,A          ; Save character
ANI 0x0F         ; Mask to 4 bits (high chars)
ADD A           ; Index * 2
; Jump table lookup and call
```

Character Range Handling:

- **0x00-0x1F**: C0 control characters (32 entries)
- **0xB0-0xBF**: High control characters (16 entries)
- **Other ranges**: Standard character processing

Video Control System

Hardware video control with software interface:

Function	Control Code	Implementation
Video On	ETX (1F)	<code>enable_video_off_control()</code>
Video Off	STX (1E)	<code>disable_video_off_control()</code>
Timeout Control	10 minutes	Automatic hardware control

Memory and Buffer Management

Character Buffers:

- Keyboard input buffer with atomic operations
- Printer output buffer (optional 2000 characters)
- Circular buffer system for tab stops
- Display memory with attribute separation

Flow Control:

- XON/XOFF software handshaking
- DTR/RTS hardware handshaking
- Character transmission masking

- Buffer overflow protection